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Course code : CSA0401

Course Name : Operating Systems

(28)

Analytical
Problem
Solving

1. CPU Scheduling - Waiting Time & Turnaround Time analysis.

Arrival Time - 0

Process	Burst Time
P ₁	10
P ₂	4
P ₃	6
P ₄	2

- Calculate WT & TAT for each process using SJF
- Calculate average WT and average TAT
- Interpret whether SJF improves CPU utilization compared to FCFS for this workload.

2. Round Robin Scheduling - Time Quantum Impact.

Process	BT
P ₁	20
P ₂	5
P ₃	13
P ₄	8

Let the Time quantum = 4ms.

- construct the Gantt chart for Round Robin
- calculate WT and TAT
- Analyse how changing the quantum to 10ms would affect system performance (CPU utilisation and context switching).

3. Deadlock - Safety check using Banker's Algorithm

A system has 3 resource types (A, B, C)

A = 10, B = 5, C = 7

Allocation

Process	A	B	C
P ₁	0	1	0
P ₂	2	0	0
P ₃	3	0	2
P ₄	2	1	1
P ₅	0	0	2

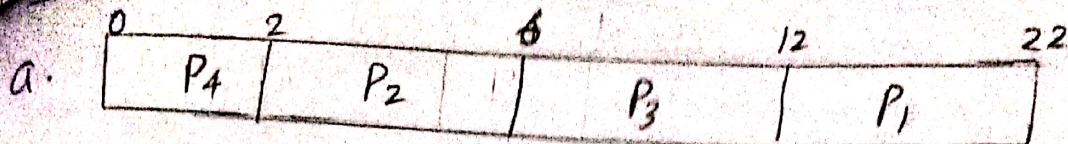
Maximum

Process	A	B	C
P ₁	7	5	3
P ₂	3	2	2
P ₃	9	0	2
P ₄	4	2	2
P ₅	5	3	3

- calculate the Need Matrix.
- Determine if the system is in safe state using Banker's Algorithm.
- Provide a valid safe sequence if it exists.

4. D,

Grantt chart for SJF



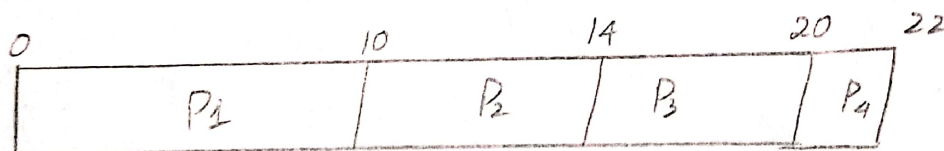
Process	BT	TAT = CT-AT	WT = TAT-BT
P ₁	10	22	12
P ₂	4	6	2
P ₃	6	12	6
P ₄	2	2	0

23
6
16
1.5
120
4

b. average WT = $(12 + 2 + 6 + 0) / 4 = 20 / 4 = 5$

average TAT = $(22 + 6 + 12 + 2) / 4 = 42 / 4 = 10.5$

c. FCFS - Grantt chart



Process	BT	TAT	WT
P ₁	10	10	0
P ₂	4	14	10
P ₃	6	20	14
P ₄	2	22	20

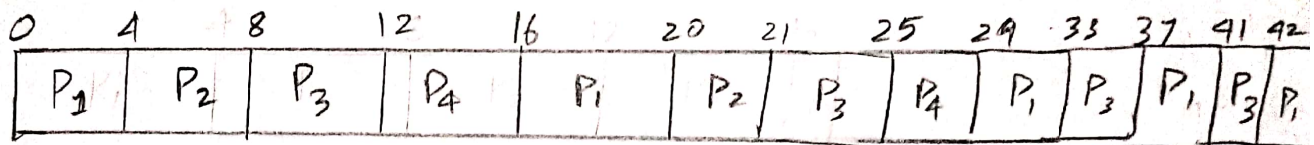
Avg WT = $(0 + 10 + 14 + 20) / 4 = 44 / 4 = 11$

Avg TAT = $(10 + 14 + 20 + 22) / 4 = 66 / 4 = 16.5$

Comparing FCFS and SJF the average waiting time and the average turn around time is comparatively higher ~~than the~~ for FCFS than for SJF.

The minimum time is performed for better CPU utilization.
So, SJF has better CPU utilization.

2. a. Gantt chart for Round Robin.



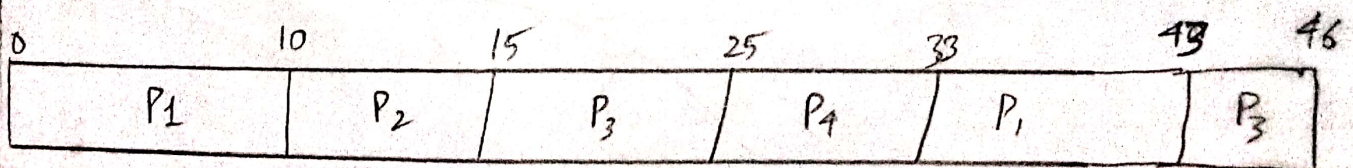
b.

Process	BT	TAT = CT - AT	WT = TAT - BT
P ₁	20	46	26
P ₂	5	21	16
P ₃	13	42	29
P ₄	8	29	21

$$\text{avg WT} = 23$$

$$\text{avg TAT} = 34.5$$

c. If Time quantum = 10 ms



Process	BT	TAT = CT - AT	WT = TAT - BT
P ₁	20	43	23
P ₂	5	15	10
P ₃	13	46	33
P ₄	8	33	25

$$\text{Avg WT} = (23 + 10 + 33 + 25) / 4 = 22.75$$

$$\text{Avg TAT} = (43 + 15 + 46 + 33) / 4 = 34.25$$

3. Available = (3, 3, 2)

a) Need Matrix = Maximum - Allocation

Process	A	B	C	
P ₁	7	4	3	✓
P ₂	1	2	2	✓
P ₃	6	0	0	✓
P ₄	2	1	1	✗✓
P ₅	5	3	1	✓

b) $P_2 \leftarrow \text{Available} \quad (2, 1, 1) \leftarrow (3, 3, 2)$

(i) $(3, 3, 2) + \text{Allocation}(P_2) = (4, 5, 4) \Rightarrow \text{New available}$
 $(5, 3, 2)$

(ii) $(5, 3, 2) > P_4$

$(5, 3, 2) + \text{Allocation}(P_4) = (7, 4, 3)$

$$\text{iii) } (7, 4, 3) + (0, 1, 0) = (7, 5, 3) \Rightarrow \text{new available.}$$

P_1

$$\text{iv) } (7, 5, 3) + P_3(3, 0, 2) = (10, 5, 5) \Rightarrow \text{new available.}$$

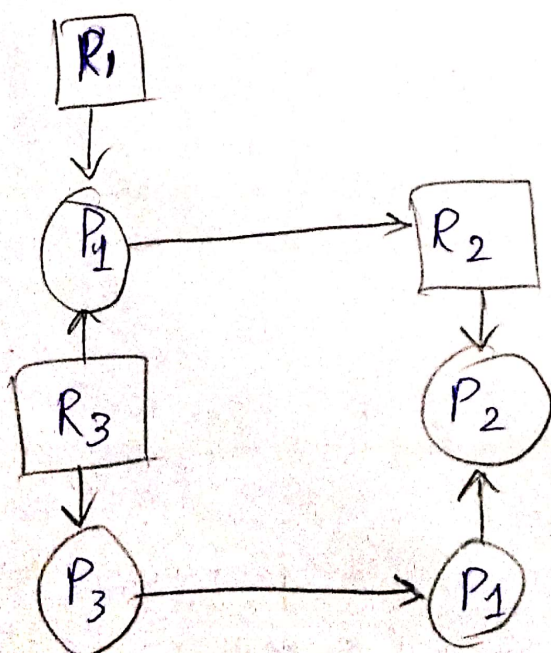
$$\text{v) } (10, 5, 5) + P_5(\overset{0, 0, 2}{\cancel{5, 3, 1}}) = (10, 5, 7) \Rightarrow \text{new available.}$$

c. P_2, P_4, P_1, P_3, P_5

4. A system has the following resource allocations.

- P_1 is holding R_1 & waiting for R_2 .
- P_2 is holding R_2 & waiting for R_3 .
- P_3 is holding R_3 & waiting for R_1 .

a. Draw the Resource Allocation Graph (RAG)



$$P_1 \rightarrow R_2 \rightarrow P_2 \rightarrow R_3 \rightarrow P_3 \rightarrow R_1 \rightarrow P_1$$

Since, there is a cycle in the Resource Allocation Graph.,
The system is in deadlock.

Deadlock status:

- Yes, the system is in deadlock.
- Each process is holding one resource and waiting for another, creating a circular wait.
- Since none of the process can proceed, the system remains permanently blocked.

c. Recommended Strategy - Recovery:

- Use deadlock recovery by terminating or restarting one of the processes to releasing its resource.
- Once R_3 is released, P_2 can continue, followed by P_1 .
- This method is effective because it breaks the circular wait and allows the remaining processes to complete their execution.

5. A system has the following process behaviours,

- Each process spends 20% time in CPU and 80% waiting for I/O.
- Degree of multiprogramming = 4 processes.

a. calculate CPU utilization ($n=4$)

$$\text{CPU utilization} = 1 - (0.8)^4$$

$$(0.8)^4 = 0.4096$$

$$= 1 - 0.4096 = 0.5904$$

$$\text{CPU utilization} = 0.5904 \times 100 = 59.04\%$$

b. calculate CPU utilization ($n=6$)

$$\text{CPU utilization} = 1 - (0.8)^6$$

$$(0.8)^6 = 0.262144$$

$$\text{CPU utilization} = 1 - 0.262144 = 0.737856$$

$$73.79\%$$

c. Interpretation

For $n=4$: CPU utilization = 59.04%

For $n=6$: CPU utilization = 73.79%

Increasing the degree of multi